STRUCTURAL DYNAMICS RESEARCH

Traffic Organization and Bridge Deck Response

Traffic organization and nominal concrete stress cycles

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Research question

How does traffic organization affect local deck stress-cycle demand when the vehicle fleet stays fixed?

Computational model

26,520

MITC4 bending/shear plate elements

Coupled steel beams and vertical supports
Approximately 79,600 algebraic coordinates
Linear equivalent model in MATLAB

Study design

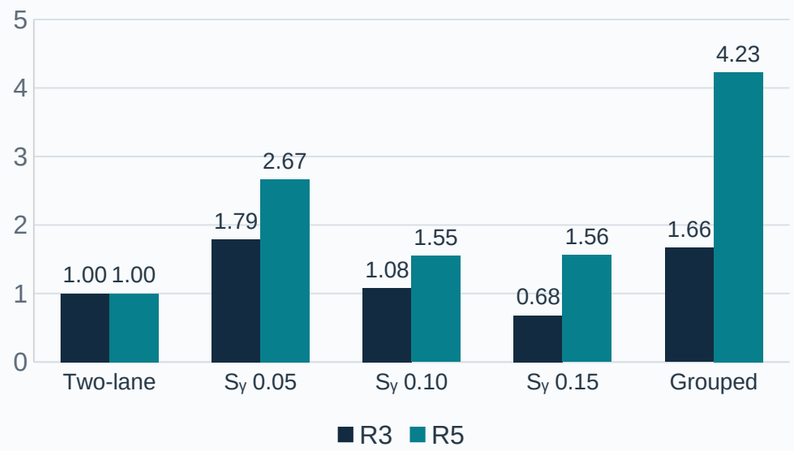
22 selected traffic scenarios use the same 240-vehicle fleet, including 20 heavy vehicles.

The comparison varies longitudinal spacing, lateral spread and vehicle ordering, with two-lane and grouped-traffic references.

Traffic comparison

At d = 40 m, the fleet and passage duration match across the displayed protocols. Ratios compare maximum fixed-channel indices to two-lane traffic.

Relative cumulative index (two-lane = 1)



Spread cases use uniform vehicle ordering. The maxima refer to fixed probes and may occur at different locations. Grouping changes the overall traffic organization.

95.53%

Minimum peak-range coverage across all 22 traffic scenarios, using both normal-stress directions.

Response coverage

Coverage compares the largest saved rainflow range with the interior plate-field temporal range. The lower longitudinal/transverse ratio defines each case.

All 22 cases reach at least 95.53%.

Section recovery

Nominal reinforcement sections cover the saved observation coordinates.

952 section variants and 3,808 independent unit-resultant checks give a maximum residual of 1.1×10^{-14}.

Interpretation

R_m = max I_m(case) / max I_m(reference)
$I_m = \Sigma\, n_i\, (\Delta\sigma_i / 1\ \mathrm{MPa})^m$

The indices screen nominal stress cycles. Peak-range coverage does not prove cumulative-index convergence. Experimental validation and material fatigue calibration remain open.

Individual work: model reconstruction, implementation, verification, simulation review and documentation. Professor Kai-Chun Chang supplied the bridge drawings.

Preliminary numerical case study. Private framework and source drawings excluded. Experimental validation and absolute fatigue life remain open.